

2005级本科生《信号与系统》期中测验

1、请说明该微分方程所描述的系统是否为线性时不变系统：

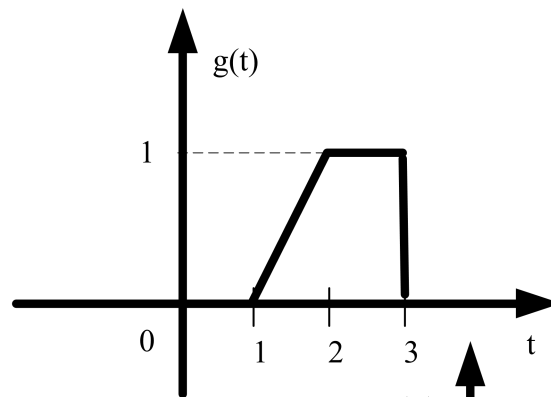
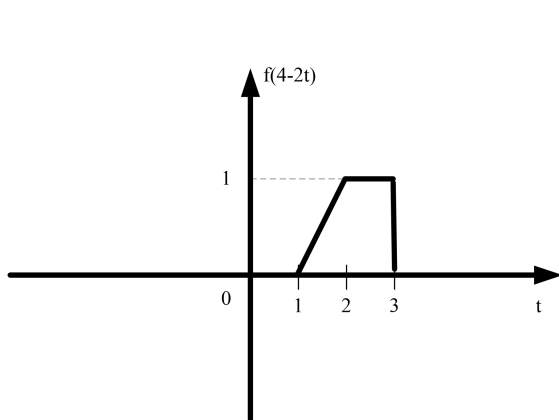
$$\frac{d}{dt}r(t) + r(t) = e(t) + 1$$

解答：不满足齐次性且系数是常数

非线性、非时变系统

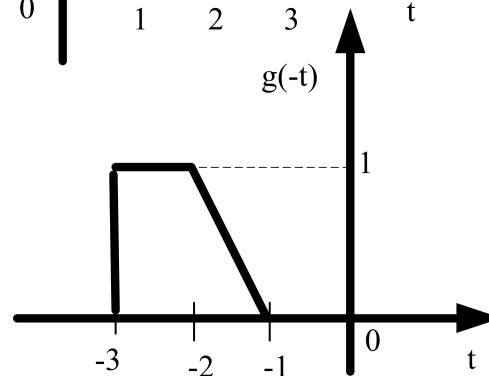
2、已知 $f(4-2t)$ 的波形如下图所示。请画出 $f(t)$ 的图形

解答：

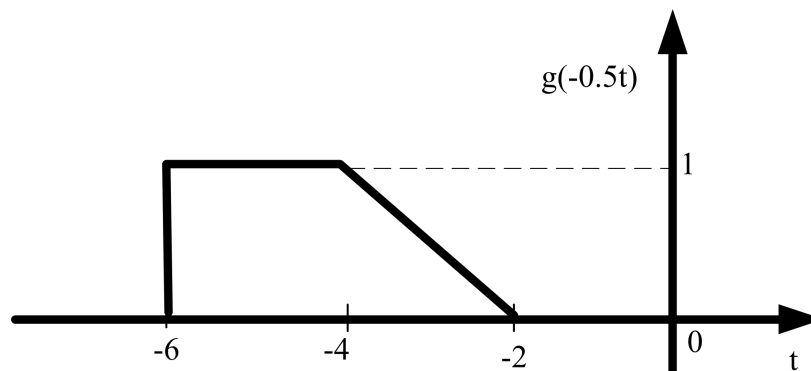
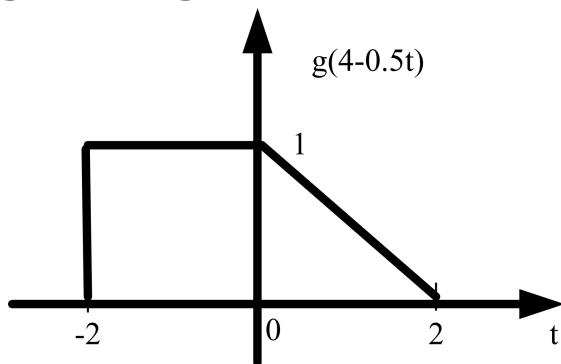


$$f(4-2t)=g(t) \quad \text{令: } 4-2t=x$$

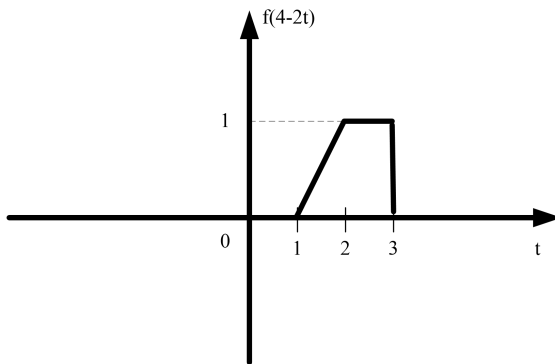
$$t=2-0.5x \quad f(x)=g(2-0.5x)$$



$$g(t) \rightarrow g(2-0.5t)$$



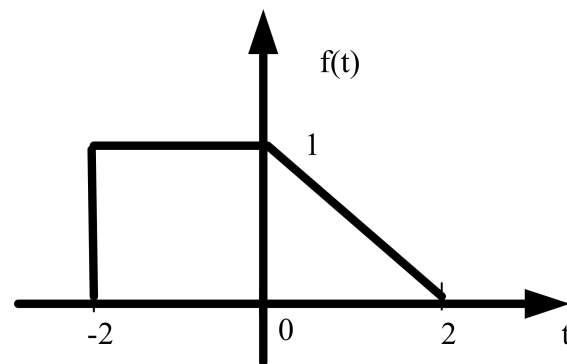
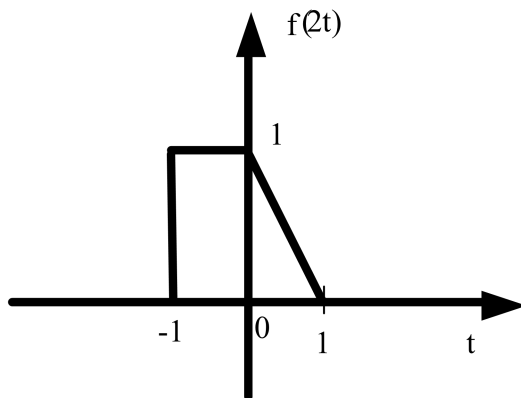
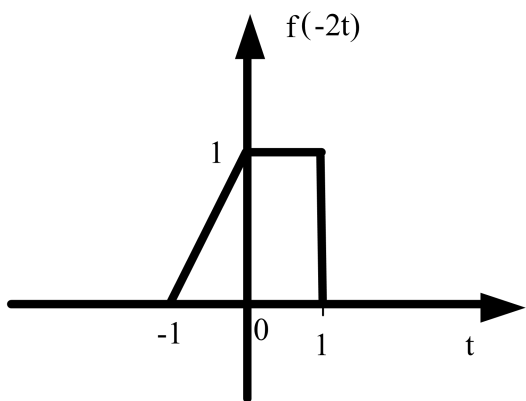
2、已知 $f(4-2t)$ 的波形如下图所示。请画出 $f(t)$ 的图形



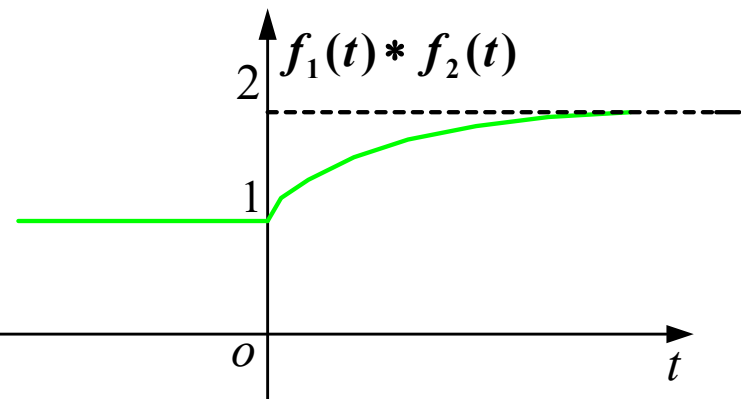
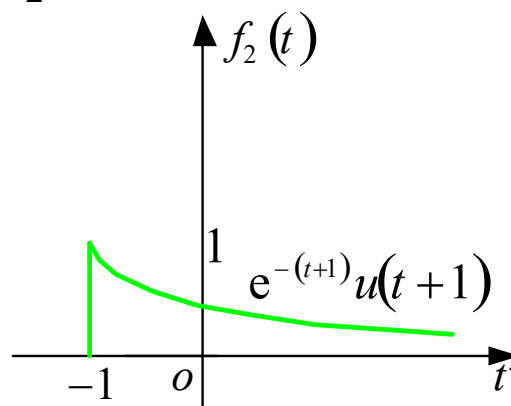
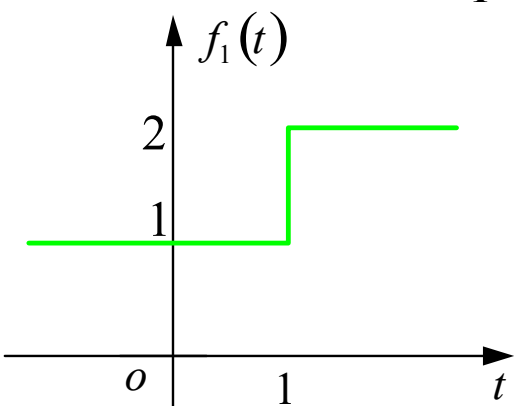
解答:

$$f(4-2t) \leftarrow f(-2t) \leftarrow f(-2t) \leftarrow f(2t) \leftarrow f(t)$$

$$f(4-2t) \rightarrow f(-2t) \rightarrow f(-2t) \rightarrow f(2t) \rightarrow f(t)$$



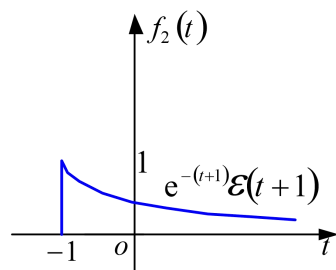
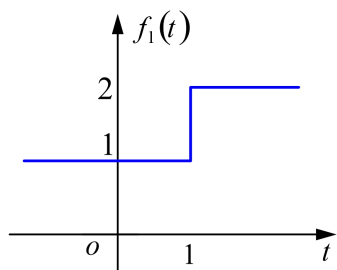
3 计算卷积 $f_1(t) * f_2(t)$, 并画出波形。



解答

$$\begin{aligned}
 s(t) &= f_1(t) * f_2(t) \\
 &= [1 + \varepsilon(t-1)] * e^{-(t+1)} \varepsilon(t+1) \\
 &= 1 * e^{-(t+1)} \varepsilon(t+1) + \varepsilon(t-1) * e^{-(t+1)} \varepsilon(t+1) \\
 &= \int_{-\infty}^{+\infty} e^{-(\tau+1)} \varepsilon(\tau+1) d\tau + \frac{d\varepsilon(t-1)}{dt} * \int_{-\infty}^t e^{-(\tau+1)} \varepsilon(\tau+1) d\tau \\
 &= \int_{-1}^{+\infty} e^{-(\tau+1)} d\tau + \delta(t-1) * \int_{-1}^t e^{-(\tau+1)} d\tau \\
 &= 1 + \int_{-1}^{t-1} e^{-(\tau+1)} d\tau = 1 + (1 - e^{-t})\varepsilon(t)
 \end{aligned}$$

3、计算并画出 $f(t) = f_1(t) * f_2(t)$ 的波形图

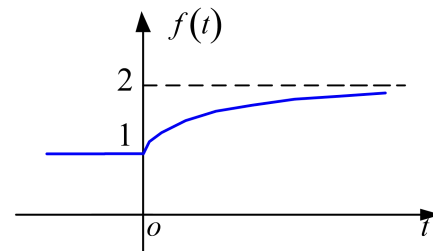


$$\begin{aligned} f(t) &= f_1(t) * f_2(t) \\ &= f_1(t+1) * f_2(t-1) \\ &= (1 + \varepsilon(t)) * e^{-t} \varepsilon(t) \end{aligned}$$

$$f_1(t) * f_2(t) \leftrightarrow F_1(j\omega) \cdot F_2(j\omega)$$

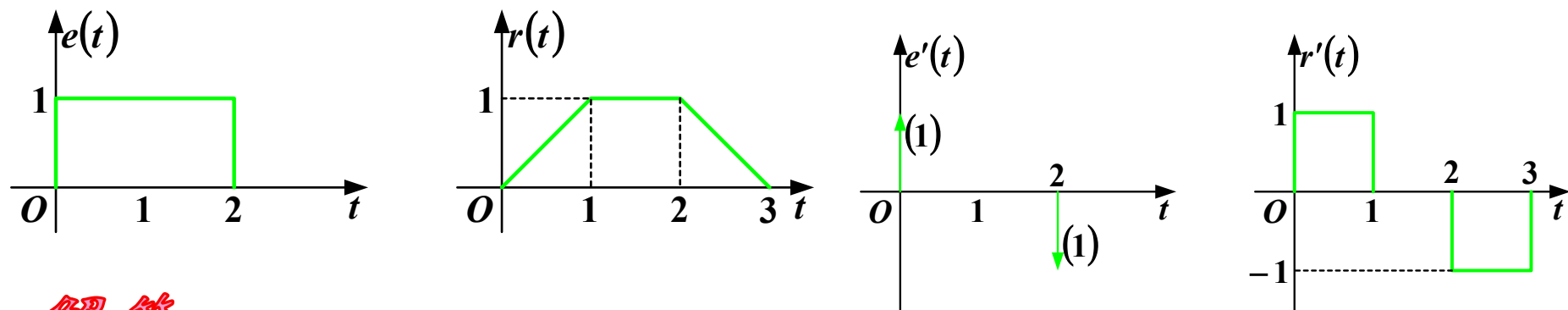
$$F(j\omega) = [2\pi\delta(\omega) + \pi\delta(\omega) + \frac{1}{j\omega}] \bullet \frac{1}{j\omega + 1}$$

$$F(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega) + \frac{1}{j\omega} - \frac{1}{j\omega + 1}$$



$$f(t) = 1 + \varepsilon(t) - e^{-t} \varepsilon(t)$$

4、已知线性时不变系统的一对激励和响应波形如下图所示，求该系统对激励的 $e_1(t) = \sin \pi t [\varepsilon(t) - \varepsilon(t-1)]$ 零状态响应。



解答

$$h(t) = \varepsilon(t) - \varepsilon(t-1)$$

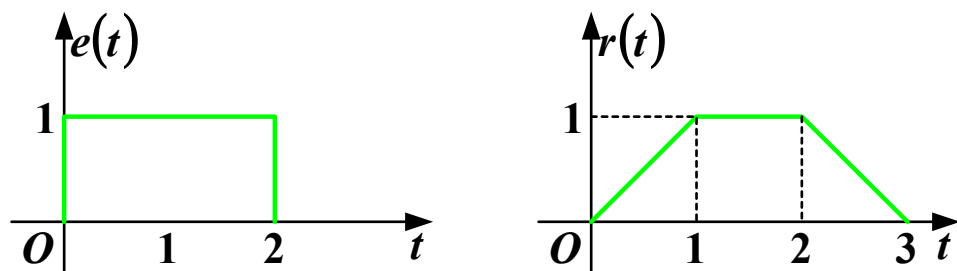
$$e'(t) = \delta(t) - \delta(t-2) \Leftrightarrow 1 - e^{-2s}$$

$$r'(t) = [\varepsilon(t) - \varepsilon(t-1)] - [\varepsilon(t-2) - \varepsilon(t-3)]$$

$$\Leftrightarrow \frac{1}{s}(1 - e^{-s}) - \frac{1}{s}(e^{-2s} - e^{-3s}) = \frac{1}{s}(1 - e^{-s})(1 - e^{-2s})$$

$$H(s) = \frac{1}{s}(1 - e^{-s})$$

4、已知线性时不变系统的一对激励和响应波形如下图所示，求该系统对激励的 $e_1(t) = \sin \pi t [\varepsilon(t) - \varepsilon(t-1)]$ 零状态响应。



解答

$$h(t) = \varepsilon(t) - \varepsilon(t-1) \quad H(s) = \frac{1}{s}(1 - e^{-s})$$

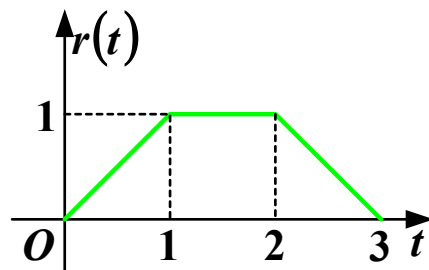
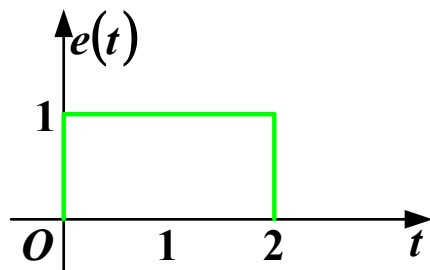
$$e_1(t) = \sin \pi t [\varepsilon(t) - \varepsilon(t-1)] = \sin \pi t \varepsilon(t) + \sin \pi(t-1) \varepsilon(t-1)$$

$$E_1(s) = \frac{\pi}{s^2 + \pi^2} (1 + e^{-s})$$

$$R_1(s) = \frac{\pi}{s^2 + \pi^2} (1 + e^{-s}) \bullet \frac{1}{s} (1 - e^{-s}) = \frac{\pi}{s^2 + \pi^2} \bullet \frac{1}{s} \bullet (1 - e^{-2s})$$

$$= \left[\frac{1}{\pi} \frac{1}{s} - \frac{1}{\pi} \frac{s}{s^2 + \pi^2} \right] (1 - e^{-2s})$$

4、已知线性时不变系统的一对激励和响应波形如下图所示，求该系统对激励的 $e_1(t) = \sin \pi t [\varepsilon(t) - \varepsilon(t-1)]$ 零状态响应。



解答

$$R_1(s) = \left[\frac{1}{\pi} \frac{1}{s} - \frac{1}{\pi} \frac{s}{s^2 + \pi^2} \right] (1 - e^{-2s})$$

$$r_1(t) = \frac{1}{\pi} [\varepsilon(t) - \cos \pi t \varepsilon(t)] - \frac{1}{\pi} [\varepsilon(t-2) - \cos \pi(t-2) \varepsilon(t-2)]$$

$$r_1(t) = \frac{1}{\pi} (1 - \cos \pi t) [\varepsilon(t) - \varepsilon(t-2)]$$

5: 已知一线性系统 $\frac{d^2 r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = 2\frac{de(t)}{dt} + 6e(t)$

$r(0_-) = 2, \quad r'(0_-) = 0, \quad e(t) = \varepsilon(t)$ 求系统的全响应,

并指出零输入响应, 零状态响应, 自由响应, 强迫响应。

一、求零输入响应 $H(s) = \frac{2s + 6}{s^2 + 3s + 2}$

$\lambda_1 = -1, \lambda_2 = -2 \quad r_{zi}(t) = C_1 e^{-t} + C_2 e^{-2t}$

$C_1 = 4, C_2 = -2$

$r_{zi}(t) = 4e^{-t} - 2e^{-2t} \quad t \geq 0$

5: 已知一线性系统 $\frac{d^2 r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = 2\frac{de(t)}{dt} + 6e(t)$

$r(0_-) = 2, \quad r'(0_-) = 0, \quad e(t) = \varepsilon(t)$ 求系统的全响应,

并指出零输入响应, 零状态响应, 自由响应, 强迫响应。

一、零输入响应 $r_{zi}(t) = (4e^{-t} - 2e^{-2t})\varepsilon(t)$

二、求零状态响应

$$E(s) = \frac{1}{s} \quad H(s) = \frac{2s + 6}{s^2 + 3s + 2}$$

$$R_{zs}(s) = H(s) \bullet E(s) = \frac{2s + 6}{s(s + 1)(s + 2)} = \frac{3}{s} - \frac{4}{s + 1} + \frac{1}{s + 2}$$

$$r_{zs}(t) = (3 - 4e^{-t} + e^{-2t})\varepsilon(t)$$

5: 已知一线性系统 $\frac{d^2 r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = 2\frac{de(t)}{dt} + 6e(t)$

$r(0_-) = 2, \quad r'(0_-) = 0, \quad e(t) = \varepsilon(t)$ 求系统的全响应,

并指出零输入响应, 零状态响应, 自由响应, 强迫响应。

零输入响应: $r_{zi}(t) = (4e^{-t} - 2e^{-2t})\varepsilon(t)$
零状态响应: $r_{zs}(t) = (3 - 4e^{-t} + e^{-2t})\varepsilon(t)$

全响应: $r(t) = (3 - e^{-2t})\varepsilon(t) = r_{zi}(t) + r_{zs}(t)$

自由分量: $-e^{-2t}\varepsilon(t)$

暂态分量 $-e^{-2t}\varepsilon(t)$

强迫分量: $3\varepsilon(t)$

稳态分量 $3\varepsilon(t)$

5: 已知一线性系统 $\frac{d^2 r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = 2\frac{de(t)}{dt} + 6e(t)$

$r(0_-) = 2, \quad r'(0_-) = 0, \quad e(t) = \varepsilon(t)$ 求系统的全响应,

并指出零输入响应, 零状态响应, 自由响应, 强迫响应。

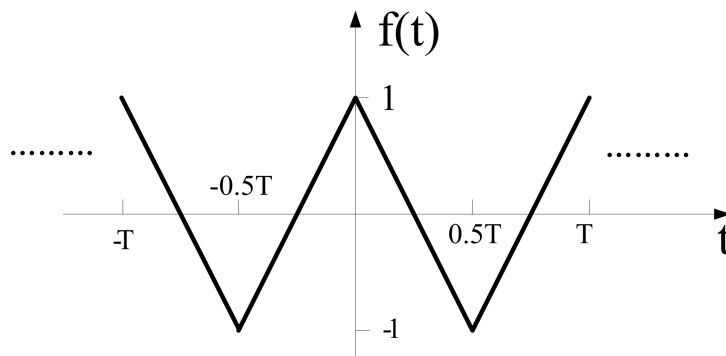
$$\left\{ \begin{array}{l} \text{零输入响应:} \quad r_{zi}(t) = (4e^{-t} - 2e^{-2t})\varepsilon(t) \\ \text{零状态响应:} \quad r_{zs}(t) = (3 - 4e^{-t} + e^{-2t})\varepsilon(t) \end{array} \right.$$

$$H(p) = \frac{2p + 6}{p^2 + 3p + 3} = \frac{4}{p + 1} - \frac{2}{p + 2}$$

$$h(t) = 4e^{-t}\varepsilon(t) - 2e^{-2t}\varepsilon(t)$$

$$\begin{aligned} r_{zs}(t) &= h(t) * e(t) = [4e^{-t}\varepsilon(t) - 2e^{-2t}\varepsilon(t)] * \varepsilon(t) \\ &= (3 - 4e^{-t} + e^{-2t})\varepsilon(t) \end{aligned}$$

6、判断图示信号 $f(t)$ 的傅立叶级数所包含的分量（6分）；



解答：

信号 $f(t)$ 是偶函数,又是奇谐函数

奇谐余弦分量

7、(8分)已知信号 $e(t)$ 的傅里叶变换 $E(j\omega)$,

$$E(j\omega) = (2 - |\omega|)[\varepsilon(\omega + 2) - \varepsilon(\omega - 2)]$$

求 $\int_{-\infty}^{+\infty} e(t) dt$ 的值。

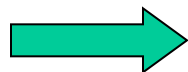
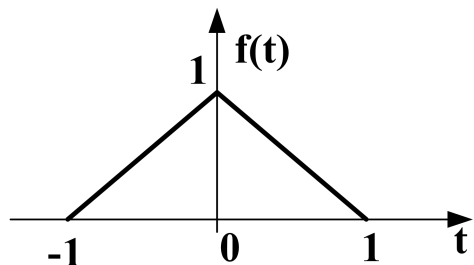
解答: $e(t) \Leftrightarrow E(j\omega)$

$$\int_{-\infty}^{+\infty} e(t) e^{-j\omega t} dt = E(j\omega) \quad \text{傅里叶正变换}$$

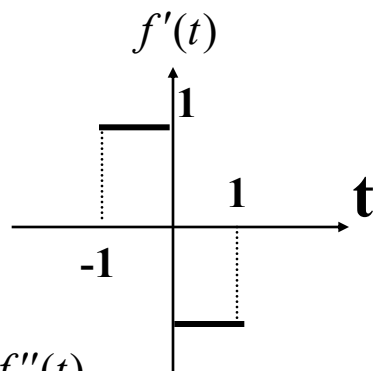
$$\int_{-\infty}^{+\infty} e(t) dt = E(0) = 2$$

8、求图信号的傅里叶变换

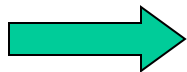
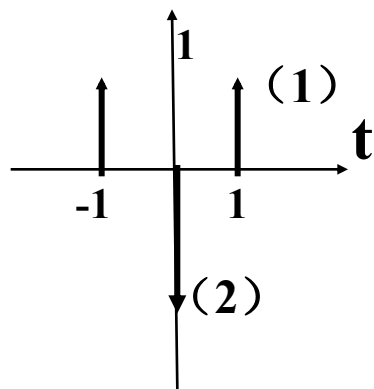
$$F(j\omega) = \frac{G(j\omega)}{j\omega} + \pi [f(\infty) + f(-\infty)]\delta(\omega)$$



$$\frac{2 \cos \omega - 2}{(j\omega)^2} = \frac{2 - 2 \cos \omega}{\omega^2}$$



$$\frac{2 \cos \omega - 2}{j\omega}$$



$$e^{j\omega} - 2 + e^{-j\omega} = 2 \cos \omega - 2$$

9、已知带宽为B 的信号 $F[f(t)] = F(j\omega)$

求信号 $f(6-2t)$ 的傅里叶变换及其带宽。

解： $f(t) \xrightarrow{\text{延时}} f(t-t_0) \xrightarrow{\text{尺度变换}} f(at-t_0)$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ F(j\omega) & F(j\omega)e^{-j\omega t_0} & \frac{1}{|a|} F(j\frac{\omega}{a})e^{-j\frac{\omega}{a}t_0} \end{array}$$

另： $f(t) \xrightarrow{\text{尺度变换}} f(at) \xrightarrow{\text{延时}} f(a(t-\frac{t_0}{a}))$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \frac{1}{|a|} F(j\frac{\omega}{a}) & & \frac{1}{|a|} F(j\frac{\omega}{a})e^{-j\frac{\omega}{a}t_0} \end{array}$$

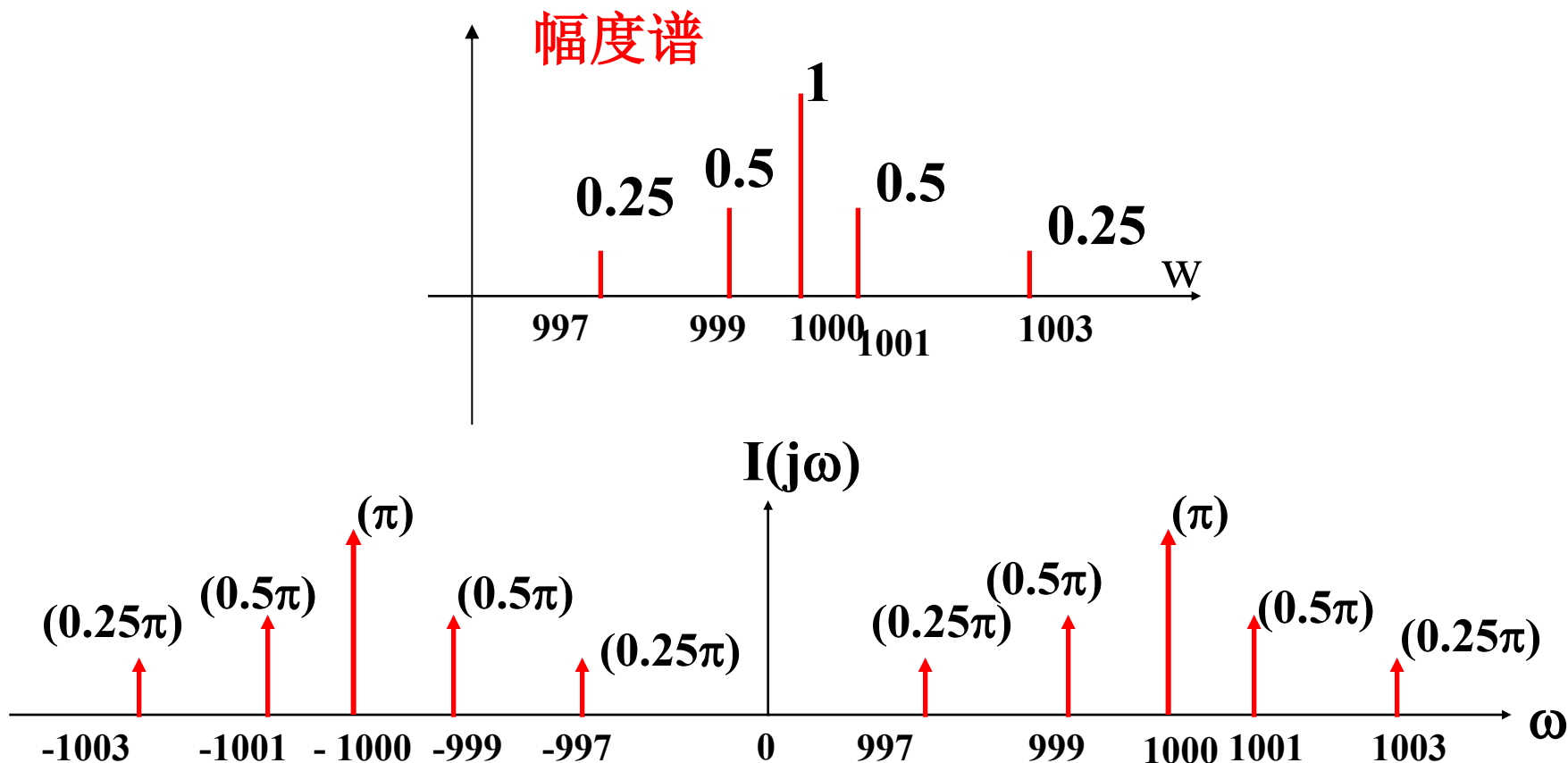
$$F[f(t)] = F(j\omega) \leftrightarrow F[f(-2t)] = \frac{1}{2} F(-j\frac{\omega}{2})$$

$$\leftrightarrow F[f(-2t+6)] = \frac{1}{2} e^{-3j\omega} F(-j\frac{\omega}{2}) \quad \text{带宽} 2B$$

10、已知幅信号电流 $i(t) = (1 + \cos t + \frac{1}{2} \cos 3t) \cos(1000t)$

画出此信号的频谱图，并计算其在电阻 1Ω 上的载波功率、最大功率、平均功率。

$$i(t) = \cos 1000t + \frac{1}{2}(\cos 1001t + \cos 999t) + \frac{1}{4}(\cos 1003t + \cos 997t)A$$



10、已知幅信号电流 $i(t) = (1 + \cos t + \frac{1}{2} \cos 3t) \cos(1000t)$

画出此信号的频谱图，并计算其在电阻 1Ω 上的载波功率、最大功率、平均功率。

$$i(t) = \cos 1000t + \frac{1}{2}(\cos 1001t + \cos 999t) + \frac{1}{4}(\cos 1003t + \cos 997t)A$$

载波功率为 $P_C = \frac{1}{2}W$

最大功率为 $P_{\max} = \frac{1}{2}(1+1+\frac{1}{2})^2 W = \frac{25}{8}W = 3.125W$

平均功率

$$\bar{P} = \left\{ \frac{1}{2} + \frac{1}{2} \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 + \left(\frac{1}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] \right\} W = \frac{13}{16} W = 0.8125W$$

11、已知某理想带通滤波器的频率响应特性为：

$$H(j\omega) = [\varepsilon(\omega + 1002) - \varepsilon(\omega + 998) + \varepsilon(\omega - 998) - \varepsilon(\omega - 1002)] e^{-j2\omega}$$

①计算其冲激响应 $h(t)$

②该系统是否物理可实现？为什么？

解：①傅里叶反变换

$$h(t) = \frac{4}{\pi} \text{Sa}(2(t-2)) \cos(1000(t-2))$$

②不满足

$$\int_{-\infty}^{\infty} \frac{|\ln |H(j\omega)| |}{1 + \omega^2} d\omega < \infty$$

非物理可实现

12、①求题10调幅信号通过题11理想带通滤波器的输出响应 $r(t)$

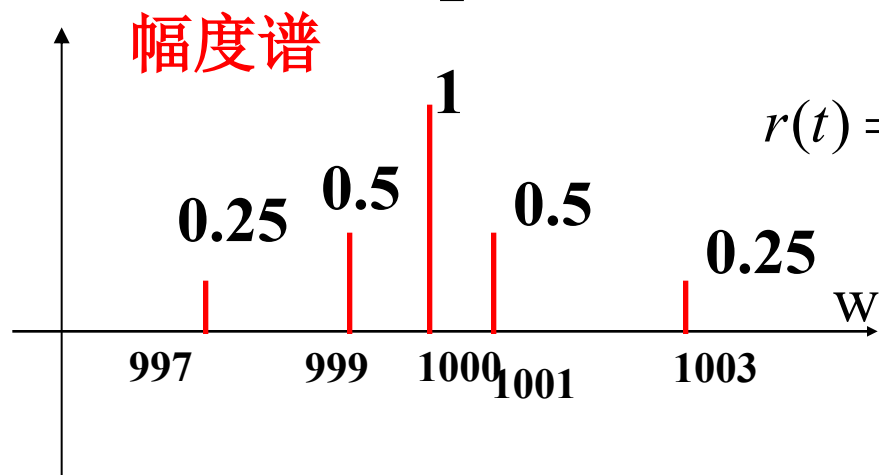
②并与题10原信号 $i(t)$ 相比较，输出信号 $r(t)$

是否存在失真？若存在失真，请说明是哪种或哪几种类型的失真？

$$i(t) = (1 + \cos t + \frac{1}{2} \cos 3t) \cos(1000t)$$

$$H(j\omega) = [\varepsilon(\omega + 1002) - \varepsilon(\omega + 998) + \varepsilon(\omega - 998) - \varepsilon(\omega - 1002)] e^{-j2\omega}$$

$$i(t) = \cos 1000t + \frac{1}{2}(\cos 1001t + \cos 999t) + \frac{1}{4}(\cos 1003t + \cos 997t)A$$



$$r(t) = [1 + \cos(t - 2)] \cos 1000(t - 2)A$$

存在幅度失真

$$r(t) = \cos 1000(t - 2) + \frac{1}{2}[\cos 1001(t - 2) + \cos 999(t - 2)]A$$